ITA 0522 - Computer vision

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1) Blister - pack inspection pipeline

1. Image acquisition:- CCD/ CMOS camera captures blister pack images

2. Image sensing :- camera reflected light into electrical signals

3. Illumination :- LED lighting provides uniform brightness.

4. Optical components :- Lens focuses light onto the camera sensor

5. Image formation :- form the tablet image on sensor

6. Sampling :- converts the image into distinct pixels

7. Quantization :- Assigns intensity values (0-255) to Pixels

8. Image enhancement :- assign numericals intensity values to each pixel. as 0-255 for 8 bit

9. Defect detection :- Image processing techniques identify missing tablets, cracks and colour changes

10. Final inspection :- The system identifies defective blister packs and automatically rejects them

2) Ceramic tile image enhancement

Intensities 100, 110, 120, 140, 185.

a) Contrast stretching

$$S = \frac{r - r_{min}}{r_{max} - r_{min}} \times 255$$

Input	Calculation	Output
100	(100-100)/85 x255	0
110	(110-100)/85x255	30
120	(120-100)/85x255	60
140	(140-100)/85 x255	120
185	(185-100)/85x255	255

b) original mean

$$\frac{100+110+120+140+185}{5} \Rightarrow \frac{655}{5}$$

$$\Rightarrow \boxed{131}$$

Enhanced mean $\Rightarrow \frac{0+30+60+120+255}{5} \Rightarrow 465/5$

$$\boxed{93}$$

c) Additional operations

1. Median filtering
   * Removes salt and pepper noise
   * preserves important edges

2. Unsharp masking
   * Enhances fine details
   * Makes fault track more visible

3. a) Histogram Equilization.

Given :- L=4 . total pixels = 64.

formula = s_k = round $[(L-1) \times CDF]$

$$P(n_k) = \frac{n_k}{64}$$

Intensity	frequency	Probability	CDF	Output
0	36	36/64 = 0.562	0.5625	2
1	16	16/64 = 0.8125	0.8125	2
2	8	8/64 = 0.125	0.9375	3
3	4	4/64 = 0.0625	1	3

Final mapping

0 -> 2 , 1 -> 2 , 2 -> 3 , 3 -> 3

b) Leaf disease image processing

1. Image acquisition : collect leaf images
2. Illumination correction : Gamma correction improves brightness.
3. Noise removal : Median filter removes noise
4. Blur reduction : Unsharp masking.
5. Background segmentation : HSV Seperation leaf
6. Resizing : Resize images to 224 x 224

7. Normalization: Scale pixel values between 0 and 1

8. Colour Enhancement :- improves colour visibility

9. Dataset prep : separate healthy and disease image

10. classification: CNN identifies leaf diseases

4. Weed inspection using mean and sharpening filter.

Given image : $\begin{bmatrix} 20 & 30 & 20 \\ 30 & 70 & 30 \\ 20 & 30 & 20 \end{bmatrix}$

$$\text{Mean} = \frac{20+30+20+30+70+30+20+30+20}{9}$$

$$\frac{270}{9}$$

$$\boxed{\text{Mean} = 30}$$

Smoothed centre intensity = 30

Step 2 :- Sharpening mask

Given Mask : $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

$$\Rightarrow (0\times20) + (-30) + 0 + (-30) + (5\times70) + (-30) - 30$$

$$\Rightarrow 350 - 120$$

$$\boxed{230}$$

Step3:- Difference.

Difference = original - Smooth

=> 70 - 30

=> 40.

Mean filtering	Sharpening
Reduces noise	Enhances edges
Smooth the image	Highlight fine details
Reduce local intensity variation.	Increase local contrast

5) Solar panel inspection using feature processing technique

1. Image acquisition: Capture solar panel image using a drone at different altitudes and viewing angles
2. Preprocessing:- Convert images into grayscale and remove noise
3. Harris corner detection:- Detect important corners and junctions in solar panels and identification plates

4. Corner selection :- Selects strong corners that are useful for identifying panel structure

5. Hough transfor : Detects straight lines and panel boundaries

6. Line detection :- Identifies rectangular shapes and alignment of solar panels

7. SIFT feature extraction: extracts distinctive key points and descriptors from images

8. Feature matching :- Matches SIFT features between images captured at different scales and angles

9. PCA : Reduces feature dimensions and removes redundant information

10. Final identification :- Combine matched feature to recognize the same solar panel and its identification plate